

COLUMBUS COUNTY AP, NC, USA

WMO: 720274

Lat: 34.273N

Lon: 78.715W

Elev: 98

StdP: 14.64

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 93799

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 24.9	(c) 28.2	(d) 2.8	(e) 6.4	(f) 34.8	(g) 8.5	(h) 8.5	(i) 36.6	(j) 18.0	(k) 63.9	(l) 15.9	(m) 63.8	(n) 2.4	(o) 0	(p) 0.352

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	94.8	76.7	92.0	76.4	90.4	75.7	80.3	88.4	79.1	86.8	78.1	85.7	6.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.7	149.4	83.7	77.2	141.9	82.4	76.2	137.5	81.7	43.8	88.4	42.7	86.7	41.6	86.3	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 15.1	(b) 12.4	(c) 10.8	(e) 18.9	(f) 98.4	(g) 3.5	(h) 2.1	(i) 16.4	(j) 99.9	(k) 14.3	(l) 101.1	(m) 12.3	(n) 102.3	(o) 9.8	(p) 103.8		
(4) 15.1	(5) 12.4	(6) 10.8	(7) 17.1	(8) 81.7	(9) 3.5	(10) 1.2	(11) 14.6	(12) 82.5	(13) 12.5	(14) 83.2	(15) 10.6	(16) 83.9	(17) 8.1	(18) 84.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	64.0	45.8	48.5	54.8	63.2	70.9	78.2	81.0	79.9	74.9	65.2	54.5	50.0	(6)
(7)		DBStd	14.46	10.18	9.34	9.01	7.54	6.24	4.85	3.73	3.67	4.85	7.92	8.30	9.49	(7)
(8)		HDD50	565	208	130	56	5	0	0	0	0	0	2	44	120	(8)
(9)		HDD65	2438	598	467	333	120	22	0	0	0	2	97	327	472	(9)
(10)		CDD50	5673	76	89	206	402	646	846	960	926	748	473	180	121	(10)
(11)		CDD65	2073	2	6	17	68	203	397	495	461	300	103	13	8	(11)
(12)		CDH74	18171	8	38	149	547	1649	3670	4957	4187	2179	688	72	27	(12)
(13)		CDH80	6946	0	2	18	109	514	1524	2190	1713	734	137	4	1	(13)
(14)	Wind	WSAvg	3.8	3.8	4.4	4.8	5.1	4.2	3.6	3.6	2.9	3.4	3.2	3.2	3.5	(14)
(15)	Precipitation	PrecAvg	53.3	4.0	3.5	3.9	3.3	3.9	5.2	5.6	6.4	6.6	3.7	3.4	3.3	(15)
(16)		PrecMax	68.4	7.8	8.1	6.5	6.2	7.4	9.5	10.8	14.0	20.2	8.0	7.9	6.8	(16)
(17)		PrecMin	33.2	1.0	1.1	0.9	0.6	1.1	3.0	2.8	1.9	1.8	0.1	0.4	1.0	(17)
(18)		PrecStd	8.5	2.0	1.7	1.6	1.5	1.7	1.9	2.0	2.9	4.4	2.3	2.2	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.7	79.3	82.4	86.8	93.1	96.9	97.4	96.8	91.6	87.6	79.9	77.5	(19)
(20)			MCWB	64.4	64.5	63.1	67.1	72.6	77.9	78.8	77.1	73.1	73.2	67.4	67.0	(20)
(21)		2%	DB	71.7	73.2	78.5	83.1	88.5	93.3	95.0	93.0	89.7	83.8	76.1	73.0	(21)
(22)			MCWB	63.3	63.0	62.8	66.0	70.9	76.1	77.1	77.4	74.4	71.3	65.4	65.5	(22)
(23)		5%	DB	67.7	69.1	74.0	80.4	85.7	90.9	92.3	90.7	87.6	80.9	72.4	70.1	(23)
(24)			MCWB	61.3	59.9	60.4	65.5	70.6	75.2	77.1	76.7	72.8	69.3	63.9	64.6	(24)
(25)		10%	DB	63.2	64.3	70.6	77.0	82.3	88.2	90.4	88.7	84.5	77.9	69.6	66.2	(25)
(26)			MCWB	56.9	56.9	59.3	63.6	69.0	73.8	76.6	76.1	72.7	68.7	62.9	62.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.8	67.4	67.9	74.0	76.1	81.5	82.4	82.0	78.5	77.3	71.3	70.8	(27)
(28)			MCDWB	70.9	73.9	75.8	79.4	84.1	92.0	92.8	88.5	85.2	80.9	76.5	74.5	(28)
(29)		2%	WB	65.0	65.6	66.1	71.3	74.1	79.1	80.4	80.2	77.5	75.4	69.1	68.5	(29)
(30)			MCDWB	69.1	72.4	73.3	77.3	82.2	87.9	89.5	86.6	84.0	80.2	72.2	71.6	(30)
(31)		5%	WB	61.9	62.4	64.1	68.8	72.8	77.8	79.1	79.0	76.2	73.5	66.6	65.3	(31)
(32)			MCDWB	66.8	66.5	70.5	75.0	80.9	86.0	87.5	85.8	82.6	78.0	70.4	68.4	(32)
(33)		10%	WB	58.3	58.9	62.1	66.8	71.6	76.6	78.1	78.0	75.0	71.3	63.9	62.2	(33)
(34)			MCDWB	62.5	63.8	68.2	73.6	79.1	84.2	86.2	84.8	81.1	75.6	67.8	65.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	19.7	20.6	22.0	22.2	20.3	18.5	18.1	16.8	17.2	19.4	21.2	19.3	(35)
(36)			MCDBR	24.8	23.8	25.6	25.0	22.5	20.5	19.8	19.0	20.4	20.3	23.0	21.0	(36)
(37)			MCWBR	17.2	14.7	12.5	11.3	8.3	7.3	6.9	6.7	7.6	9.5	13.3	14.1	(37)
(38)		5% WB	MCDWB	21.7	22.2	21.4	19.4	18.6	18.4	18.4	16.8	15.9	14.8	18.8	18.8	(38)
(39)			MCWBR	17.8	16.6	13.2	10.9	7.7	7.4	7.3	6.6	6.7	8.0	14.6	14.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.323	0.333	0.369	0.401	0.465	0.493	0.537	0.512	0.433	0.397	0.350	0.323	(40)	
(41)		taud	2.500	2.469	2.394	2.295	2.165	2.131	2.027	2.091	2.308	2.370	2.463	2.534	(41)	
(42)		Ebn at Noon	281	291	287	281	263	255	243	247	263	263	268	275	(42)	
(43)		Edh at Noon	27	31	36	42	48	49	55	50	39	34	28	25	(43)	
(44)	All-Sky Solar Radiation	RadAvg	864	1074	1426	1793	1951	2026	1932	1732	1478	1236	954	762	(44)	
(45)		RadStd	51	106	115	117	145	126	122	94	115	139	75	61	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)		+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+306	+218

Nomenclature: See separate page